

The Obidi Convention and the Theory of Entropicity (ToE)

Scholarly Edition

Chapter 1

Executive Summary

Purpose and Scope

This document presents a comprehensive scholarly reconstruction and expansion of John Onimisi Obidi's introductory supplement to the Theory of Entropicity (ToE). The objective is to transform the original conceptual exposition into a mathematically rigorous and pedagogically self-contained reference suitable for researchers, graduate students, and advanced readers in theoretical physics, information geometry, gravitation, and complex systems.

The Theory of Entropicity proposes that spacetime geometry, gravitation, and the dynamical laws governing physical systems emerge from deeper informational and entropic structures. The theory seeks to provide a unified framework in which entropy is elevated from its traditional role as a statistical descriptor to that of a fundamental dynamical entity underlying physical reality.

The present reconstruction preserves the conceptual content of the original supplement while significantly expanding its mathematical clarity, explanatory depth, and logical structure.

The Central Thesis of the Theory of Entropicity

The central proposition of ToE is that the geometry of spacetime and the phenomenon of gravitation are not fundamental features of nature. Rather, they emerge from a deeper informational substrate governed by entropic principles.

This viewpoint is motivated by several developments in modern theoretical physics.

The discovery of black-hole entropy established a profound connection between gravitation and thermodynamics. The work of Jacobson demonstrated that Einstein's field equations may be derived from thermodynamic relations imposed on local causal horizons. Verlinde subsequently proposed that gravitational attraction may be interpreted as an emergent entropic force rather than a fundamental interaction.

The Theory of Entropicity extends these ideas further by proposing that informational geometry itself possesses physical significance and that entropy governs the emergence of both information geometry and spacetime geometry.

In symbolic form, the conceptual hierarchy of ToE may be represented as

Entropy $\longrightarrow$ Information $\longrightarrow$ Geometry $\longrightarrow$ Gravity $\longrightarrow$ Physical Dynamics.

According to this viewpoint, Einsteinian gravitation emerges as a limiting description of a more general entropic field theory.

Information Geometry as a Foundation

A major conceptual pillar of ToE is the recognition that information possesses intrinsic geometric structure.

The theory draws heavily upon two established geometrical frameworks:

Fisher–Rao Geometry

The Fisher–Rao metric provides a Riemannian geometry on the space of probability distributions.

For a family of distributions $p(x|\theta)$, the Fisher information metric is

$$g_{ij}(\theta) = \int p(x|\theta) \frac{\partial \ln p(x|\theta)}{\partial \theta^i} \frac{\partial \ln p(x|\theta)}{\partial \theta^j} dx.$$

This metric quantifies statistical distinguishability between neighboring probability distributions.

Fubini–Study Geometry

Quantum states likewise possess a natural geometry.

The Fubini–Study metric provides the canonical metric on the projective Hilbert space of pure quantum states.

Consequently, both classical and quantum information admit rigorous geometrical descriptions.

The Theory of Entropicity adopts these geometrical structures as foundational ingredients.

Hybrid Manifold Architecture System (HMAS)

ToE introduces a generalized manifold consisting of multiple geometric sectors.

The Hybrid Manifold Architecture System (HMAS) contains three distinct sectors:

$$J \in \{FR, FS, L\}.$$

These correspond respectively to:

FR = Fisher–Rao Sector,

FS = Fubini–Study Sector,

L = Lorentzian Sector.

Each sector possesses its own metric tensor

$$g_{\mu\nu}^{(J)}.$$

The total geometric structure is obtained by combining the individual sector contributions through the Obidi Addition Rule,

$$g_{\mu\nu} = g_{\mu\nu}^{(FR)} + g_{\mu\nu}^{(FS)} + g_{\mu\nu}^{(L)}.$$

This decomposition constitutes one of the defining mathematical features of the theory.

The Obidi Convention

To accommodate multiple geometric sectors simultaneously, ToE introduces an extended tensor notation known as the Obidi Convention.

Every tensor index consists of two components:

1. A primary coordinate index.
2. A secondary sector index.

Thus tensor components are written in the form

$$T_{(\mu)J},$$

where

$$\mu = 0, 1, 2, 3$$

is the conventional spacetime index and

$$J \in \{FR, FS, L\}$$

identifies the sector.

This convention allows a unified treatment of classical information geometry, quantum information geometry, and spacetime geometry within a single tensor framework.

Obidi Calculus

The theory introduces a collection of algebraic rules collectively known as Obidi Calculus.

Obidi Addition Rule

Physical tensors are decomposed into sectoral contributions:

$$T_\mu = T_{(\mu)FR} + T_{(\mu)FS} + T_{(\mu)L}.$$

Obidi Multiplication Rule

Multiplicative structures factorize across sectors:

$$Z_{HMAS} = \prod_J Z^{(J)}.$$

Similarly,

$$|g| = |g_{FR}| |g_{FS}| |g_L|.$$

Obidi Contraction Rule

Contractions occur only between matching sectors.

Thus

$$T^{(\mu)J} S_{(\mu)K} = 0, \quad J \neq K. \quad =$$

This principle establishes orthogonality among distinct sectors.

The Entropy Field

The dynamical content of ToE is governed by an entropy scalar field

$$S(x).$$

Unlike conventional thermodynamic entropy, which is often interpreted statistically, the entropy field in ToE possesses direct dynamical significance.

The field couples to geometry and influences the evolution of the Hybrid Manifold.

The entropy field therefore plays a role analogous to that of a fundamental field in modern physics.

Local Obidi Action

The dynamics of the theory are generated by the Local Obidi Action (LOA).

For each sector J , the action is written

$$I_{LOA}^{(J)} = \int \left(R^{(J)} - \Lambda_{(J)} + \kappa_{(J)} g^{(J)\mu\nu} \nabla_\mu^{(J)} S \nabla_\nu^{(J)} S + V(S_{(J)}) \right) e^{S/k_B} \sqrt{|g^{(J)}|} d^4x.$$

The complete action is

$$I_{LOA} = \sum_J I_{LOA}^{(J)}.$$

The exponential factor

$$e^{S/k_B}$$

represents the coupling between entropy and geometry.

This term distinguishes the Obidi Action from the conventional Einstein–Hilbert action.

Obidi Field Equations

Variation of the Local Obidi Action yields the master field equations of the theory.

For each sector one obtains

$$G_{(\mu\nu)J} + \Lambda_{(J)} g_{(\mu\nu)J} = 8\pi G_{(J)} T_{(\mu\nu)J}^{ent}.$$

These equations govern the interaction between geometry and entropy throughout the Hybrid Manifold.

Together they constitute the fundamental dynamical equations of the Theory of Entropicity.

Recovery of General Relativity

A crucial consistency requirement for any proposed extension of gravitational theory is recovery of Einstein's equations in an appropriate limit.

The Theory of Entropicity satisfies this condition.

When entropy gradients vanish,

$$\nabla_\mu S = 0,$$

or become sufficiently small, the Fisher–Rao and Fubini–Study sectors decouple from the dynamics.

Only the Lorentzian sector remains active.

The Obidi Field Equation then reduces to

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

This is precisely the Einstein Field Equation of General Relativity.

Therefore General Relativity appears as the low-entropic-gradient limit of the broader entropic framework.

Scientific Significance

The Theory of Entropicity belongs to a growing family of approaches seeking to derive gravitation from deeper informational principles.

However, ToE differs from previous approaches in several important respects.

Jacobson derived Einstein's equations from local thermodynamic constraints.

Verlinde interpreted gravity as an emergent entropic force.

Padmanabhan emphasized horizon thermodynamics and entropy extremization.

Bianconi employed relative entropy and information geometry to generate modified gravitational dynamics.

The Theory of Entropicity attempts a more comprehensive unification by simultaneously incorporating:

- Classical information geometry.
- Quantum information geometry.
- Lorentzian spacetime geometry.
- Entropic field dynamics.
- Hierarchical tensor calculus.

into a single formal framework.

Summary

The Theory of Entropicity proposes that entropy constitutes the fundamental organizing principle from which information, geometry, and gravitation emerge.

To accomplish this, the theory introduces:

- The Hybrid Manifold Architecture System.
- The Obidi Convention.
- Obidi Calculus.
- The Local Obidi Action.
- The Obidi Field Equations.

Within this framework, Einsteinian gravity is recovered as a limiting case of a deeper entropic theory.

The remainder of this monograph develops these ideas in detail, beginning with a rigorous statement of the theory's mathematical foundations and culminating in the derivation of the Obidi Field Equations and their relationship to General Relativity.